INCH-POUND

MIL-C-63418 (AR)
AMENDMENT 2
7 May 1992

SUPERSEDING
AMENDMENT 1
6 April 1991

MILITARY SPECIFICATION

CANISTER SMOKE, ASSEMBLY
PARTS FOR ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-C-63418 (AR), dated 1 December 1980, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1: STANDARDS, MILITARY

Add: "MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes"

Delete: "MIL-STD-1218 - ACS Chemicals"

PAGE 2

2.1: Drawings, Army, add:

"12934724 - Alternate Canister Smoke, Assembly"

* 2.2: Publications, ASTM

Delete: "ASTM, D2905 - Statement of Number of Specifications Required to Determine the Average Quality of a Textile"

* Add the following new paragraph 2.3:

"2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3.2 Add the following:

"3.2.1 Alternate canister smoke, assembly. The assembly and components shall comply with all requirements specified on Drawing (Dwg.) 12934724 and with the requirements specified in applicable specifications."

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3.6 Add the following:

"3.6 Sealing. The alternate smoke canister assembly shall withstand a vacuum of 6.0 +/- 1.0 inches of mercury for a minimum period of 30 seconds without leakage."

* 3.7 Add the following:

"3.7 Proving ground. The smoke canister, when assembled into a projectile, fuzed, and tested as specified, shall upon ground impact functioning produce a smoke signature visible across 4000 meters of level terrain."

* 3.8 Add the following:

"3.8 Smoke mix process control. The process for weighing, mixing and dispensing the smoke mix and constituents thereof shall be established, and qualified by the first article proving ground test. Once qualified, changes to the equipment or procedures shall require prior approval of the contracting officer. Re-qualification shall also be required if in the opinion of the Government procuring activity, maintenance or repair to the mixing or dispensing mechanism effects process control parameters. The following process controls are mandatory:

a. Weigh and check weigh each constituent prior to addition to blender. The check weigh shall be a different device by a different operator or inspector.

b. Minimum mixing time shall be that used to mix the first article (qualification) batch(s).

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c. Mixing speed shall be within plus or minus 10% of that used to mix the first article (qualification) batch(s)."

* 4.3.1, Line 7 under "Quantity" delete "20" and substitute "25"

PAGE 4

* 4.3.1 Add the following after "Pad 9342877 5":

"If the container has elected to produce the alternate smoke canister assembly, then the first article sample shall consist of the following items in sample quantities as indicated:

"Part Description	Drawing	Quantity
Alternate Canister Smoke, Assembly	12934724	25
Body	12934726	5
Lid	12934725	5
Felt	12934727	5"

* 4.3.2.1, Line 2, Delete "composition and"

PAGE 5

* Delete "TABLE I" and substitute the following included "TABLE 1."

PAGE 11

* 4.4.2.5, Block "NEXT HIGHER ASSEMBLY": Delete "9331795" and substitute "9331794"

* 4.4.2.5 Delete Major Defect 101 in its entirety and substitute:

"101 Mixing process incorrect 4.4.3.1 3.8 Visual"

PAGE 12

* 4.4.2 Add the included new tables: 4.4.2.7, 4.4.2.8, 4.4.2.9, and 4.4.2.10.

(The above new tables 4.4.2.7, 4.4.2.8, 4.4.2.9, and 4.4.2.10 are identical to those in Amendment 1, except for 4.4.2.10 where only Major 101 is changed. From "Major 101", Batch formulations incorrect, 4.4.3.1, 3.2.1, 4.5.1" To "Major 101" Mixing process incorrect, 4.4.3.1, 3.8, Visual")

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Canister Smoke, Assembly and Components	1 OF 1	See below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Canister Smoke, Assembly (Dwg. 9333792)				
	Examination for Defects (except Ballistic Test)	5		3.2	4.4.2.5
	Ballistic Test	20		3.7	4.5.1
	Components Examination for Defects	5		3.2	4.4.2
	Alternate Canister Smoke, Assembly (Dwg. 12934724)				
	Examination for Defects (except Ballistic Test)	5		3.2.1	4.4.2.10
	Ballistic Test	20		3.7	4.5.1
	Components Examination for Defects	5		3.2.1	4.4.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Body (Alternate Canister Smoke)		12934726
			NEXT HIGHER ASSEMBLY 12934724

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness through base, min.		0.40%	3.2.1	Gage
102	Inside diameter of neck		0.40%	3.2.1	Gage
103	Outside diameter of base end		0.40%	3.2.1	Gage
104	Outside diameter of lip		0.40%	3.2.1	Gage
105	Wall thickness, min.		0.40%	3.2.1	Gage
106	Total length		0.40%	3.2.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Lid (Alternate Canister Smoke)	1 OF 1	12934725
			NEXT HIGHER ASSEMBLY: 12934724

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2.1	Gage
102	Outside diameter of neck		0.40%	3.2.1	Gage
103	Outside diameter		0.40%	3.2.1	Gage
Minor					
201	Thickness of lining compound		0.65%	3.2.1	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

6 (12b)

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QUALITY CONFORMANCE INSPECTION

MIL-C-63418 (AR)
AMENDMENT 2

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Felt (Alternate Canister Smoke)		12934727
			NEXT HIGHER ASSEMBLY 12934724

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter		0.65%	3.2.1	Gage
202	Thickness		0.65%	3.2.1	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

7 (12c)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Alternate Canister Smoke, Assembly		12934724
			NEXT HIGHER ASSEMBLY
			9331794

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Mixing process incorrect		4.4.3.1	3.8	Visual
102	Moisture content over max.		4.4.3.2	3.5	4.5.2
103	Canister not sealed		4.4.3.3	3.6	4.5.3
104	Length		0.40%	3.2.1	Gage
105	Any component missing or incorrectly assembled		0.40%	3.2.1	Gage
106	Weight of composition		0.40%	3.2.1	Balance
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual
202	Marking missing, misleading, or unidentifiable		0.65%	3.2.1	Visual

NOTES:

8 (12d)

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* Delete 4.4.3.1, 4.4.3.1.1, 4.4.3.1.2, 4.4.3.1.2.1, 4.4.3.1.2.2 in their entirety and substitute:

"4.4.3.1 Mixing process control. Prior to dispensing each batch (see 6.8) of smoke mix, the following process control checks shall be made:

a. Check records of weigh and check weigh of all constituents.

b. Check mixing time, minimum.

c. Check mixing speed record for entire minimum time.

d. Examine weighing and mixing area for evidence of spilled constituents in excess of 1% of batch weight. (Do not include dispensing area).

If mixing time or speed are not as required, re-mix the batch for the required minimum time at the required speed. If the weigh and check weigh was not accomplished for any constituent, or if spilled constituents are in evidence, the batch shall be rejected."

* Delete 4.4.3.1.2.3 in its entirety,

4.4.3.3 Add the following:

"4.4.3.3 Sealing. Each shift's production of smoke canister assemblies shall be tested for leakage. Acceptance for this test shall be in accordance with MIL-STD-105, Inspection Level I, AQL 1.0% with the inspection lot equal to the number of smoke canister assemblies produced by one seaming machine during one shift. Component parts selected for assembly into leak test samples shall be loaded with inert composition and marked as inert prior to seaming. The leak test samples shall not be returned to the inspection lot."

* Delete 4.5 in its entirety, and substitute:

"4.5 Methods of Inspection."

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* Delete 4.5.1 and 4.5.1.1 in their entirety and substitute:

"4.5.1 Proving ground. The sample smoke canister assemblies shall be assembled into M804, 155MM, projectiles and tested at a Government proving ground in accordance with the applicable acceptance test procedure. Projectiles shall be fuzed with an M739 series PD fuze set "PD". The projectiles shall be fired in a M185 or M199 Series cannon using a M119A1, Zone 8 charge. Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 21 degrees Celsius (70 degrees F) plus or minus 1 degree Celsius (2 degrees F) for a minimum of 24 hours prior to testing. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. An observer shall be stationed 4000 + 200 meters from the edge of the impact area to record presence of smoke signature. The test shall be conducted on level terrain and the wind velocity shall not exceed 15 knots at the time of testing."

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* Delete 4.5.1.2 in its entirety.

PAGE 17

* Delete 4.5.1.3 in its entirety.

PAGE 18

4.5.3 Add the following:

"4.5.3 Sealing. Place inert smoke canister assembly in fixture and close lid. Start vacuum. Lower canister into water. After gage shows 6.0 +/- 1" of mercury, observe canister for a minimum of 30 seconds. Raise the canister out of the water. Release vacuum. Raise the lid, remove the canister, and dry off. Set aside the canister for scrap. If the canister is a reject, then identify the seaming machine that produced the reject, and reject all canisters produced by that seaming machine since the last acceptable leak test sample."

5.2 Delete in its entirety.

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* 6.3 Delete "ARRADCOM.....07801" and substitute "ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, N.J. 07806-5000."

* 6.6 Delete in its entirety and substitute the following:

"6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the projectile is procured by the US Army Armament, Munitions and Chemical Command, two copies of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-FSA-MS and SMCAR-QAR-Q, Picatinny Arsenal, NJ 07806-5000, and Commander, ARRCOM, ATTN: AMSMC-QAD, Rock Island, IL 61299-6000."

* 6.9 Delete "ARRADCOM...07801" and substitute "ARDEC, ATTN: SMCAR-QAR-Q, Picatinny Arsenal, N.J. 07806-5000."

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* Delete 6.10 in its entirety.

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